

● EASY

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

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# Area and volume

# 02

10 answers · Rationale-rich · Teach from doc

**Q1****A****A.** 18

Choice A is correct. The volume,  $V$ , of a rectangular prism is given by the formula  $V = lwh$ , where  $l$  is the length of the base,  $w$  is the width of the base, and  $h$  is the height of the prism. It's given that each base of a right rectangular prism has a length of 19 inches and a width of 8 inches, and that the prism has a volume of 2,736 cubic inches. Substituting 19 for  $l$ , 8 for  $w$ , and 2,736 for  $V$  in the formula  $V = lwh$  gives  $2,736 = (19)(8)(h)$ , or  $2,736 = 152h$ . Dividing each side of this equation by 152 yields  $18 = h$ . Therefore, the height, in inches, of the prism is 18.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the area, in square inches, of the base of the prism, not the height, in inches, of the prism.

**Q2****C****C.** 3,025

Choice C is correct. The area  $A$ , in square centimeters ( $\text{cm}^2$ ), of a square with side length  $s$ , in cm, is given by the formula  $A = s^2$ . It's given that the square has a side length of 55 cm. Substituting 55 for  $s$  in the formula  $A = s^2$  yields  $A = 55^2$ , or  $A = 3,025$ . Therefore, the area, in  $\text{cm}^2$ , of the square is 3,025.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the perimeter, in cm, of the square, not its area, in  $\text{cm}^2$ .

Choice D is incorrect and may result from conceptual or calculation errors.

**Q3****986**

The correct answer is 986. The area,  $A$ , of a rectangle is given by  $A = lw$ , where  $l$  is the length of the rectangle and  $w$  is its width. It's given that the length of the rectangle is 34 centimeters (cm) and the width is 29 cm. Substituting 34 for  $l$  and 29 for  $w$  in the equation  $A = lw$  yields  $A = 3429$ , or  $A = 986$ . Therefore, the area, in square centimeters, of this rectangle is 986.

**Q4****A**

A. 8

Choice A is correct. It's given that the area of a square is 64 square inches. The area  $A$ , in square inches, of a square is given by the formula  $A = s^2$ , where  $s$  is the side length, in inches, of the square. Substituting 64 for  $A$  in this formula yields  $64 = s^2$ . Taking the positive square root of both sides of this equation yields  $8 = s$ . Thus, the side length, in inches, of this square is 8.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the area, in square inches, of the square, not the side length, in inches, of the square.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q5****D**

D. 52

Choice D is correct. It's given that rectangle P has an area of 72 square inches. If a rectangle with an area of 20 square inches is removed from rectangle P, the area, in square inches, of the resulting figure is  $72 - 20$ , or 52.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

**Q6****180**

The correct answer is 180. The perimeter of a polygon is equal to the sum of the lengths of the sides of the polygon. It's given that each side of the square has a length of 45. Since a square is a polygon with 4 sides, the perimeter of this square is  $45 + 45 + 45 + 45$ , or 180.

**Q7****C****C.** 42

Choice C is correct. The area,  $A$ , of a rectangle is given by the formula  $A = lw$ , where  $l$  represents the length of the rectangle and  $w$  represents its width. It's given that the rectangle has a length of 7 inches and a width of 6 inches. Substituting 7 for  $l$  and 6 for  $w$  in the formula  $A = lw$  yields  $A = 76$ , or  $A = 42$ . Thus, the area, in square inches, of the rectangle is 42.

Choice A is incorrect. This is the sum, not the product, of the length and width of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is twice the area, in square inches, of the rectangle.

**Q8****D****D.** 1,568

Choice D is correct. The area,  $A$ , of a rectangle can be found using the formula  $A = lw$ , where  $l$  represents the length and  $w$  represents the width of the rectangle. It's given that a rectangle has a length of 56 inches and a width of 28 inches. Substituting 56 for  $l$  and 28 for  $w$  in the formula  $A = lw$  yields  $A = (56)(28)$ , or  $A = 1,568$ . Therefore, the area, in square inches, of the rectangle is 1,568.

Choice A is incorrect. This is the width, in inches, not the area, in square inches, of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

**Q9****D****D.** 329

Choice D is correct. It's given that the width of this rectangle is 7 centimeters and that the length of this rectangle is 40 centimeters longer than the width. Therefore, the length of this rectangle is  $7 + 40$ , or 47, centimeters. The area of a rectangle can be found by multiplying its length and its width. Therefore the area, in square centimeters, of this rectangle is  $7 \times 47$ , or 329.

Choice A is incorrect. This is the width, in centimeters, not the area, in square centimeters, of this rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

**Q10****B****B.** 8

Choice B is correct. The perimeter of a triangle is the sum of the lengths of all three of its sides. It's given that the lengths of two sides of a triangle are 4 centimeters and 6 centimeters. Let  $x$  represent the length, in centimeters, of the third side of this triangle. The sum of the lengths, in centimeters, of all three sides of the triangle can be represented by the expression  $4 + 6 + x$ . Since it's given that the perimeter of the triangle is 18 centimeters, it follows that  $4 + 6 + x = 18$ , or  $10 + x = 18$ . Subtracting 10 from both sides of this equation yields  $x = 8$ . Therefore, the length, in centimeters, of the third side of this triangle is 8.

Choice A is incorrect. If the length of the third side of this triangle were 2 centimeters, the perimeter, in centimeters, of the triangle would be  $4 + 6 + 2$ , or 12, not 18.

Choice C is incorrect. If the length of the third side of this triangle were 10 centimeters, the perimeter, in centimeters, of the triangle would be  $4 + 6 + 10$ , or 20, not 18.

Choice D is incorrect. If the length of the third side of this triangle were 24 centimeters, the perimeter, in centimeters, of the triangle would be  $4 + 6 + 24$ , or 34, not 18.

Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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